Manual Segmentation & Spray Metrics

Complete Workflow: `manual_segment.py`

Workflow Overview

1. **Prerequisites** — `.cine` file + `config.json` from `GUI.py`
2. **Load** — video & calibration config
3. **Pre-process** — robust scaling, gain/gamma
4. **Segment** — SAM3 single-frame or full-video
5. **Refine** — brush / contour tools; static block
6. **Preview** — Play Boundary
7. **Export** — metrics CSV, boundary CSV, AVI

Required files

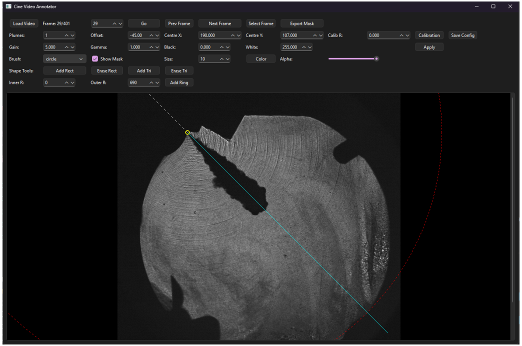
- `*.cine` — raw high-speed camera recording
- `config.json` — calibration generated by `GUI.py`

What `config.json` contains

- Injector centre (x, y) in pixels
- Rotation offset (deg)
- Inner radius (orifice exit distance, px)
- Outer radius (strip width, px)
- Number of plumes

Generate config with `GUI.py`

Open `GUI.py`, load a reference frame, mark the injector centre and radii, save `config.json` to the same directory as the `.cine` files.



Step 2 — Launch `manual_segment.py`

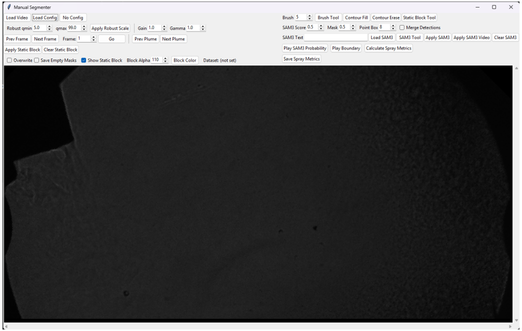
Run from a terminal in the repo root:

```
python manual_segment.py
```

Load Video
Click *Load Video* → select the `.cine` file.
The video is extracted into per-plume strip arrays stored in `plume_videos`.

Load Config
Click *Load Config* → select `config.json`.
Sets centre, rotation, inner/outer radius; strips are re-extracted and aligned automatically.

Tip: use *No Config* only for exploratory viewing — metrics will use default (zero) inner radius.



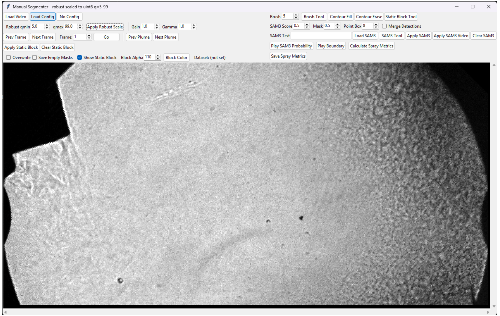
Step 3 — Pre-processing: Robust Scale + Gain/Gamma

Apply Robust Scale

- Clips the intensity range to $[q_{\min}, q_{\max}]$ percentiles and re-maps to 8-bit.
- Removes fixed hot/dead pixels that would confuse SAM3.
- Typical values: $q_{\min} = 1\%$, $q_{\max} = 99\%$.

Gain & Gamma (display + model input)

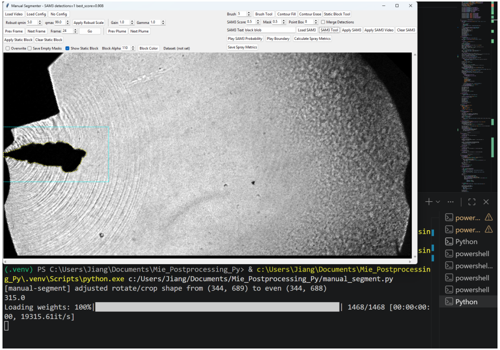
- *Gain* — linear brightness multiplier ($\times 1.0$ default).
- *Gamma* — power-law contrast ($\gamma = 1.0 =$ linear).
- Both are applied when rendering frames *and* when writing the temporary AVI sent to SAM3 Video — so the model sees the enhanced image.

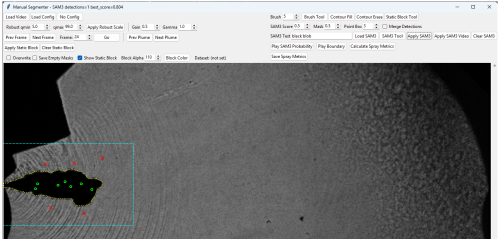


Step 4a — SAM3 Single-Frame Segmentation

1. Click **Load SAM3** (loads `sam3.pt` once).
2. Enter a text prompt in the *SAM3 Text* field e.g. `spray plume`.
3. Click **SAM3 Tool** to enable the prompt mode.
4. Optionally draw a bounding box on the canvas (left-drag) or click positive/negative points.
5. Click **Apply SAM3** — model runs on the current frame only; mask is written to `plume_masks`.

Use single-frame mode to *initialise* the mask on a representative frame before running full-video propagation.



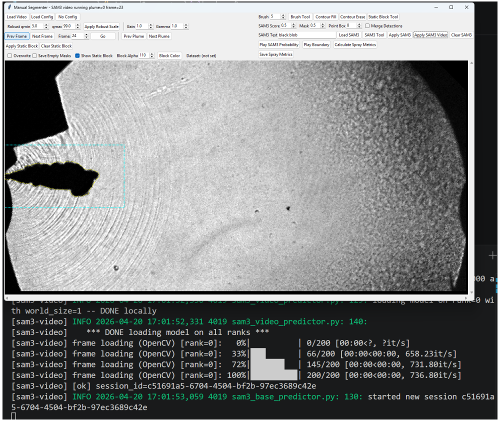


Step 4b — SAM3 Full-Video Propagation

Apply SAM3 Video

1. Ensure a text prompt and/or mask/box is set on the reference frame.
2. Click **Apply SAM3 Video**.
3. The tool writes a temporary `input.avi` (gain/gamma applied), then launches the SAM3 video predictor in a background thread.
4. Progress is printed to the terminal; the title bar updates on completion.
5. Masks for all frames are loaded into `plume_masks`.

Note: SAM3 Video uses cross-frame memory propagation — temporal consistency is prioritised over per-frame accuracy. Manual refinement (next step) compensates for difficult frames.



Step 5 — Manual Mask Refinement

Brush Tool

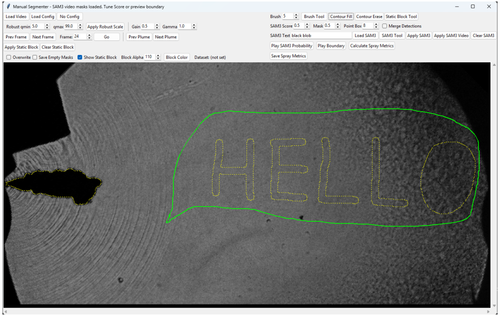
- Left-drag: *add* to mask (white).
- Right-drag: *erase* from mask.
- Adjust *Brush* size spinbox before painting.

Contour Fill / Contour Erase

- Left-click inside a closed region to flood-fill add or erase.
- Useful for filling enclosed holes or removing detached blobs.

Static Block Tool

- Draw a persistent rectangular block that is *always erased* from masks across all frames (e.g. nozzle body, reflections).
- Click *Apply Static Block* to propagate; *Clear* to remove.

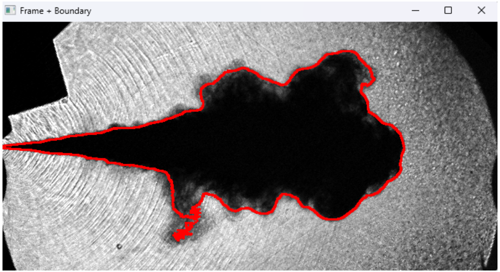


Step 6 — Preview: Play Boundary

Click **Play Boundary** to open an OpenCV window that loops through all frames with the current `plume_masks` rendered as blue boundary lines.

- The boundary is computed via morphological erosion on each mask — no separate storage, always reflects the latest edits.
- Press **Q** to close the preview window.
- Use this step to verify that manual edits propagated correctly before exporting.

Play SAM3 Probability (separate button) plays the raw SAM3 confidence map from the last *Apply SAM3 Video* run — useful for diagnosing model confidence on specific frames without overwriting edits.



Step 7a — Calculate Spray Metrics (Preview)

Click **Calculate Spray Metrics** to compute and *display* per-frame metrics from the current masks.

Key metrics derived from the binary plume envelope

- **Penetration_from_BW** — furthest occupied column (px), shifted to orifice-local $x = 0$.
- **Penetration_from_BW_Polar** — max Euclidean distance from orifice to boundary.
- **Cone_Angle_Average** — mean $\arctan(y/x)$ of boundary points in x-band.
- **Cone_Angle_Linear_Regression** — regression slope at fixed orifice intercept.
- **Area** — foreground pixel count per frame.
- **Estimated_Volume** — axisymmetric envelope integral.

Nozzle *opening* / *closing* frames are detected from the near-nozzle binary signal and marked as vertical lines on all plots.

Step 7b — Save Spray Metrics (Export)

Click **Save Spray Metrics** → choose output directory.

Three files are written:

1. `{stem}_plume{N}_metrics.csv`
   One row per frame; columns identical to `mie_single_hole_pipeline` output — directly comparable with automated BW results.
2. `{stem}_plume{N}_boundary_points.csv`
   Per-point table: `frame, side, y, x`. Same format as `save_boundary_csv` in the pipeline.
3. `{stem}_plume{N}_boundary.avi`
   Rendered boundary video (20 fps) — identical visual to Play Boundary.

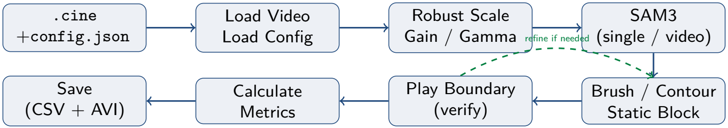
A confirmation dialog displays nozzle opening/closing frames and the output paths.

Comparing Manual vs. Automated Pipeline

Same metric definitions, different segmentation source. Both call `extract_single_plume_features` + `features_to_binarized_metrics_df` — CSV columns are byte-for-byte compatible for direct quantitative comparison.

Metric	Automated Pipeline	Manual Segment
Penetration (BW & Polar)	✓	✓
Cone Angle (avg & LR)	✓	✓
Area	✓	✓
Volume (proxy)	✓	✓
Nozzle opening / closing	✓	✓
Boundary points CSV	✓	✓
Boundary AVI	—	✓

Summary — Complete Closed Loop



Key design choices

- All edits write to a single `plume_masks` array — SAM3 and brush share the same state.
- Gain/Gamma affects both display *and* the video sent to the model.
- Metrics use orifice-local $x = 0$; `inner_radius` from config ensures alignment.

Output files

- `*_metrics.csv` — frame-wise spray metrics
- `*_boundary_points.csv` — boundary point cloud
- `*_boundary.avi` — visualisation video